

HEMASAI VATTIKUTI

☎ +91 79937 40469 ✉ hemasaihattikuti2727@gmail.com [in](#) LinkedIn [G](#)itHub [P](#)ortfolio

Summary

Final-year CSE student at VIT-AP. Interned at DRDO (Ministry of Defence) building distributed defence infrastructure with sub-10s failover. Founder of Mithra Life OS — AI productivity platform with 941 authenticated users. Stack: Python, FastAPI, PostgreSQL, LLMs, RAG.

Skills

Languages: Python, JavaScript, SQL

AI / LLMs: RAG Pipelines, Agentic AI, Vector Embeddings, Semantic Search, Gemini API, Groq API, ONNX Runtime, pgvector, Qdrant, Prompt Engineering

Backend: FastAPI, Async Python, Pydantic v2, JWT, RBAC, Rate Limiting, Uvicorn, REST APIs

Databases: PostgreSQL, MongoDB (Replica Sets, Aggregation Pipelines), Neon, Supabase, pgvector, Qdrant

DevOps : Docker • Docker Compose • Linux • Render • Nginx • Playwright • Git • Vercel • Row-Level Security

Experience

Defence Research and Development Laboratory (DRDL – DRDO)

Aug – Nov 2025

Project Intern • Backend & Distributed Systems • Hyderabad, India

- Built a high-availability backend for DAMS v3.0 by architecting a 3-node MongoDB Replica Set with automatic PRIMARY election and sub-10s failover under Scientist E, Ministry of Defence
- Built async FastAPI backend with JWT, bcrypt, RBAC, IP-rate limiting (100 req/min via slowapi), Pydantic v2 validation, and 60s TTL caching; MongoDB aggregation pipelines for inventory analytics and live DB health via `serverStatus`
- Containerised 5 services (3 DB nodes + FastAPI + Nginx/React) with Docker Compose; health monitors poll CPU/DB every 30s and auto-trigger failover on threshold breach

Orion Movers, New Zealand • [Live](#)

Jun – Jul 2026

Full-Stack Developer (Remote)

- Automated quote-to-lead pipeline: form submission triggers email notification and CRM entry simultaneously with zero manual client intervention post-launch
- Deep technical SEO (Open Graph, Twitter Card, geo/ICBM meta, canonical tags) — site ranks first page on Google for target NZ search queries within weeks of launch
- Sole developer delivering a production commercial web application using Next.js (Vercel), FastAPI (Render), Supabase, Resend API, and CRM integration.

Projects

Mithra Life OS • [Live](#) • [GitHub](#)

2025 – Present

React • FastAPI • Neon PostgreSQL • Firebase Auth • pgvector • Docker • Capacitor

- Offline-first sync engine: bidirectional last-write-wins merge, deduplicated localStorage queue, exponential backoff (30s cap with jitter), real fetch health ping every 30s
- Cut Gemini API costs via client-side classifier (regex + 4-hour TTL cache); custom Pydantic-to-Gemini `$ref` resolver; dual-model strategy (Flash/Pro) with Python fallback on outages
- Built and launched full-stack AI productivity platform with 941 authenticated Firebase users with zero marketing spend

VITAp-UniOS • [Live](#) • [GitHub](#)

2026

• FastAPI • Qdrant • FastEmbed (ONNX) • Groq Llama-3 • Playwright

- Production RAG campus assistant: 955.5K LLM tokens processed (Groq Console, last 30 days)., 240+ daily API requests across 30 consecutive days — runs within Render's 512MB RAM replacing PyTorch with FastEmbed ONNX (MiniLM-L6-v2); binaries pre-cached for zero cold-download startups
- Parallel hybrid retrieval (vector + lexical scroll) with term-frequency reranking (+0.30 title, +0.075 body); false-positive filter and campus isolation scope eliminate cross-campus hallucination entirely
- Authority-aware Playwright crawler with midnight rebuild pipeline and automated RAG evaluation suite (`evaluate_rag.py`) running score threshold and keyword checks on every nightly rebuild

Achievements

- 4th Place** — Google Cloud Hackathon (GoogleDevs Sprint 2K25) • VIT-AP • 486 participants • Mar 2025
- 7th Place** — Simverse Hackathon • VIT-AP University • 200+ participants • Jan 2026

Certifications

- AI Agents and Agentic AI with Python & Generative AI — Vanderbilt University (Coursera) • Oct 2025
- Advanced Learning Algorithms — Stanford / DeepLearning.AI • Jul 2025
- Supervised ML: Regression & Classification — Stanford / DeepLearning.AI • Apr 2025

Education

Vellore Institute of Technology – Andhra Pradesh (VIT-AP)

2023 – 2027

B.Tech, Computer Science and Engineering • Amaravati, AP

Coursework: Data Structures & Algorithms, Machine Learning, NLP, Database Management Systems, Linear Algebra